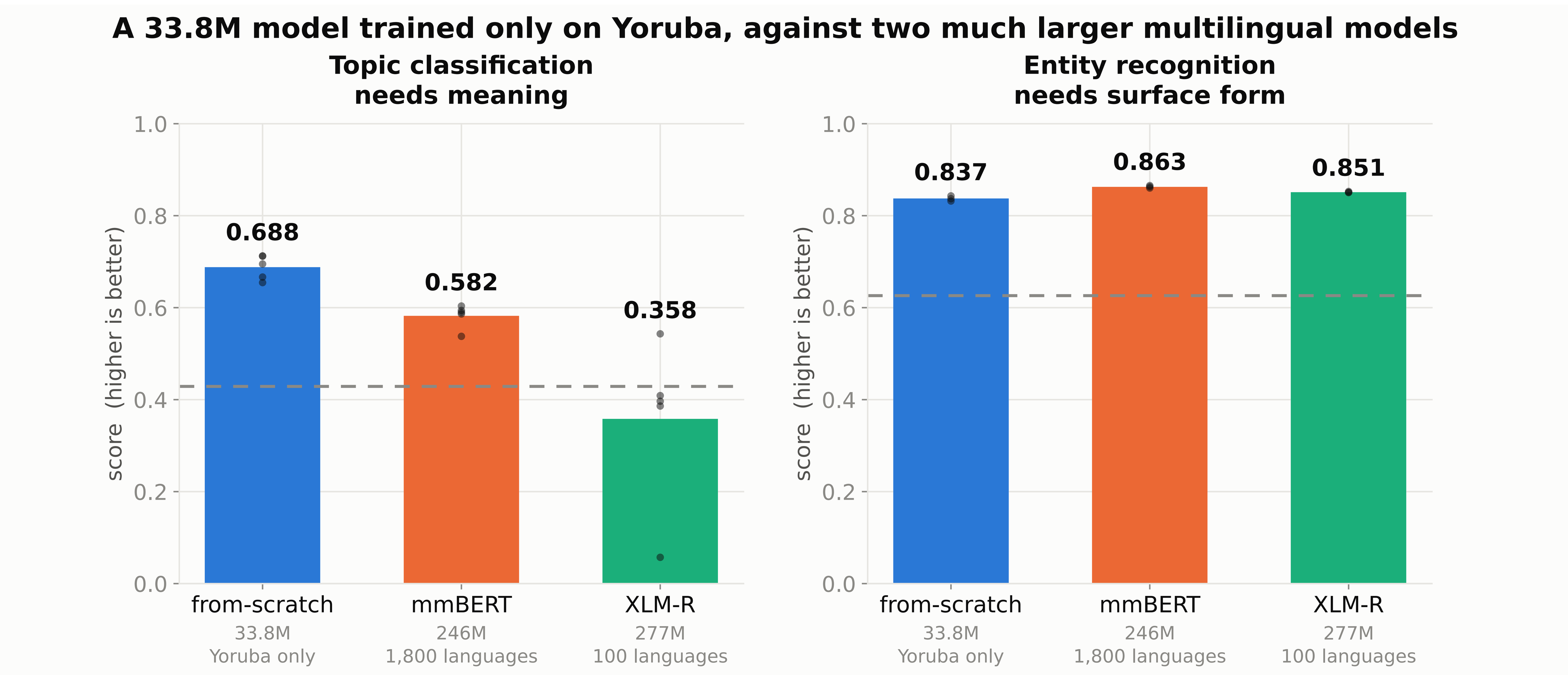
When Is It Worth Training Your Own Model?

Yoruba from scratch against multilingual transfer

TAKEAWAY **A 33.8M-parameter Yoruba model is ahead of mmBERT on topic classification; the strongest evidence points to vocabulary fit—not text scarcity—as the leverage.**

RESULT



The dashed line is what the same architecture scores with no pretraining at all: 0.429 on topic classification, 0.626 on entity recognition.
Dots are individual seeds.
XLM-R on topic classification trained in 4 of 5 seeds, 1 collapsing below chance -- a bar like that is a mixture, not a mean.
Learning rates for entity recognition were chosen on the same test items they are scored on, which inflates them; the other panel was chosen on a held-out split.

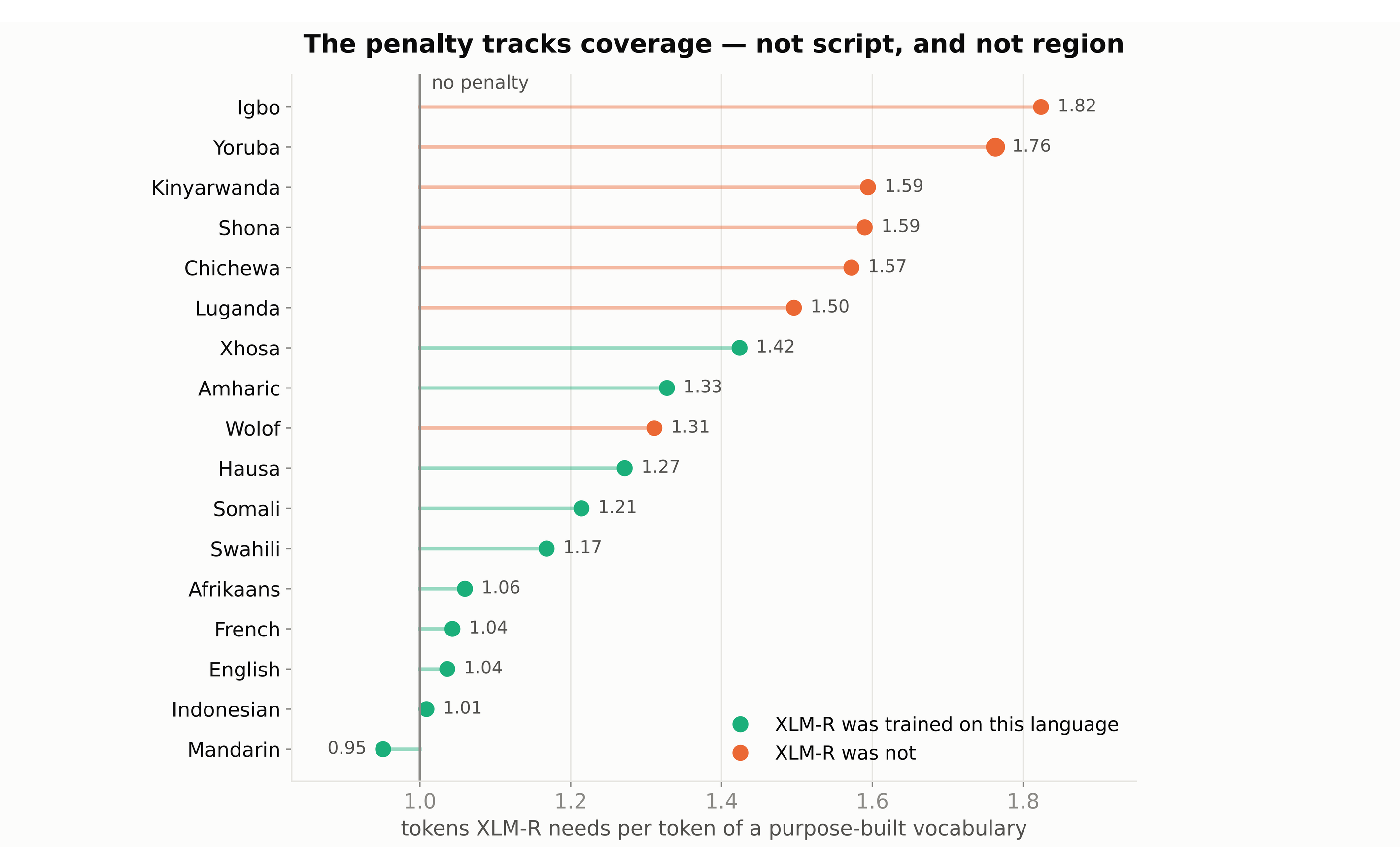
A small model, a precise claim

On SIB-200, the 33.8M from-scratch model reached 0.688 macro-F1 versus 0.582 for mmBERT (five seeds; learning rates chosen on the dev split). The 0.106 margin is much larger than seed noise, but bootstrap intervals overlap by 0.004: say “ahead,” not “beats.”

Causal vocabulary swap

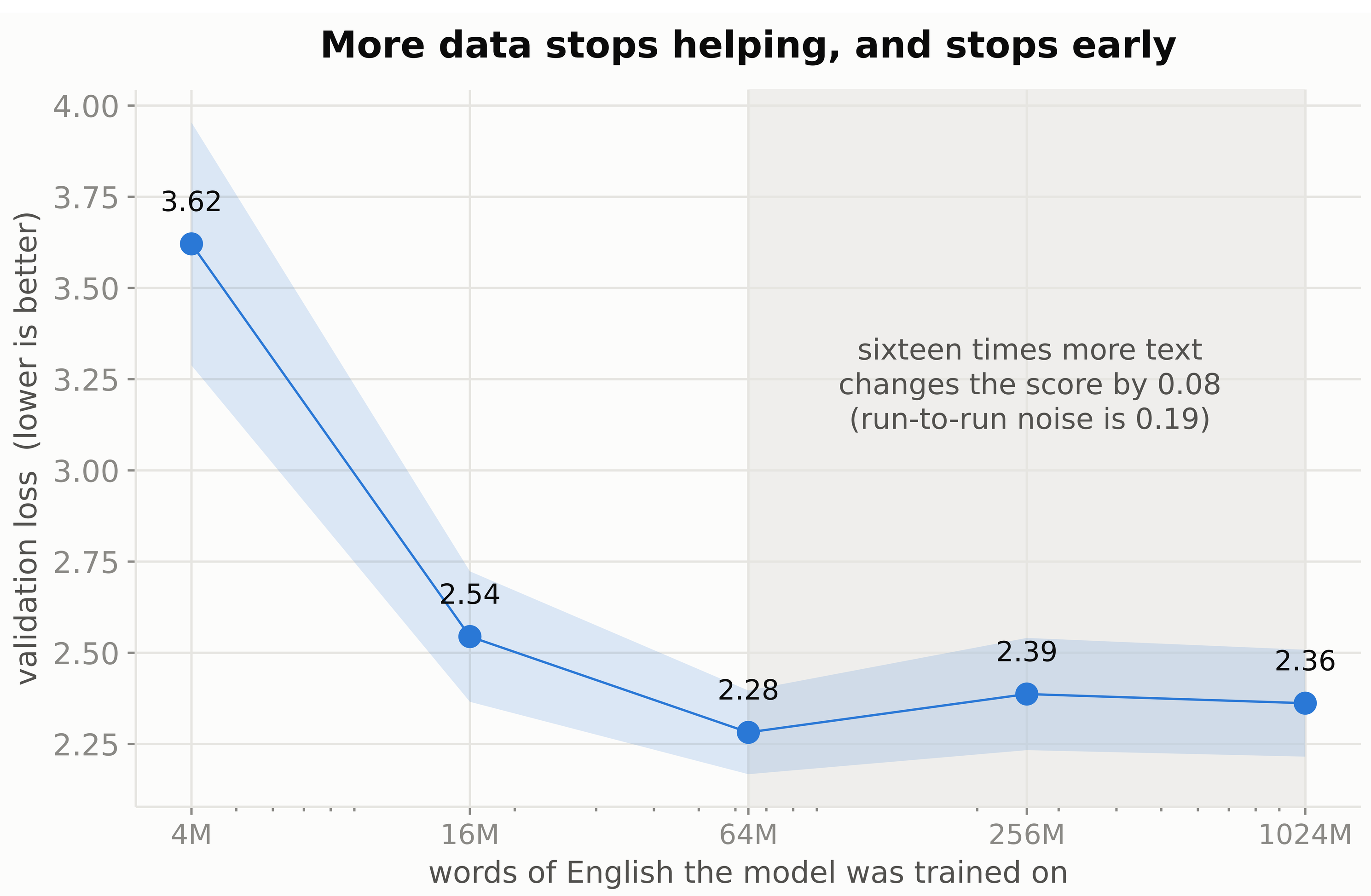
Holding architecture, Yoruba text and compute fixed, swapping only the vocabulary changed downstream scores. The 16k vocabulary led the 250k XLM-R vocabulary by +0.144 on SIB-200 and +0.061 on MasakhaNER; every seed won in both tasks (four seeds per arm, exact p=0.029).

1 · COVERAGE

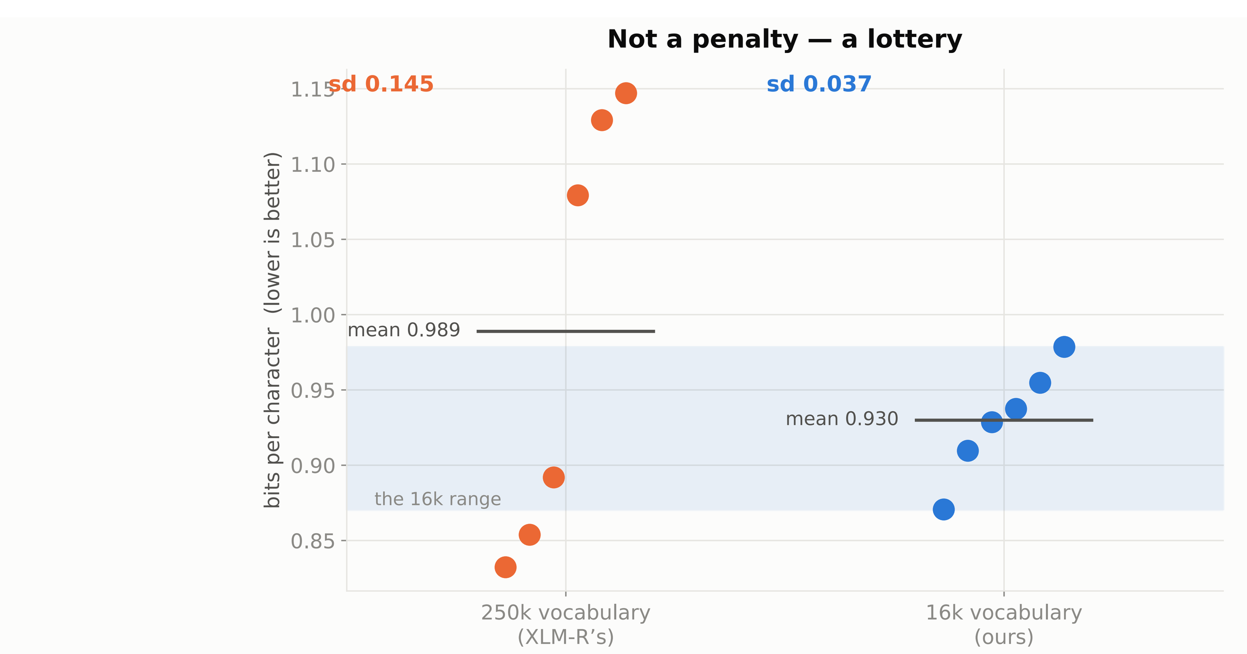


XLM-R's vocabulary costs the 10 languages it covers 1.150× on average, and the 7 it does not 1.593×.
Restricted to African languages on both sides — which rules out script and region as the explanation — 1.244× against 1.593×.
6 of the 7 uncovered languages sit above every covered one. The exception is not smoothed: Wolof at 1.31× is uncovered and below Xhosa at 1.42×.
Yoruba, the target language, ranks 2 of 17 at 1.76× — so the original figure was not a quirk of one language.

2 · DATA



3 · VARIABILITY



6 seeds a side, the sample size fixed in advance. The MEANS do not separate: +0.059 bits/char, p = 0.37, and 3 of the 6
250k runs land below the 16k median. The SPREADS do: F = 15.1, p = 0.0098. Downstream, over 4 pretraining seeds: topic 0.7× (p = 0.553), entities 8.6× (p = 0.005).